Introduction to Focus Areas – Data Science –
WS26/27 Katharina Jahn

Assignment 1

Organisation

- Deadline Nov 6, 10:00
- Only **one person** per group submits the homework; the names of all persons of the group have to be marked on the submission & in the filenames (name1, name2, name3)
- Submission format: 2 files
  - **1 notebook file** named **AssignmentX_notebook_name1_name2_name3.ipynb** contains the self-contained, documented code, in particular all packages you used; uses relative paths to our provided datasets; provides **only** a selection of output/figures that is asked for
  - **1 pdf** named **AssignmentX_notebook_name1_name2_name3.pdf** is your notebook converted to pdf

The Data

We will use the Heart Disease Data Set „The Cleveland Database“, see https://archive.ics.uci.edu/ml/datasets/Heart+Disease and also this content for describing the data source.

You find a copy of the dataset on the Whiteboard.
This database contains clinical and biomedical attributes for over 300 patients. The "goal" field refers to the presence of heart disease in a patient. It is integer-valued from 0 (no presence) to 4.

Tasks:

1. Perform an exploratory data analysis on the Cleveland heart disease dataset described above. You can reuse your code from the in-class project. Label the analysis with respect to the data analysis steps in the lecture (Initial Data Analysis 1-7)
2. Train at least three types of classifiers to diagnose heart disease based on the available data and compare them visually in a grouped bar chart with regard to accuracy, sensitivity, and specificity. Create a confusion matrix for each classifier and plot them as a heatmap.
3. Repeat the steps of the previous task, using as class labels only “heart disease vs. no heart disease”. Analyze the implications of this change. Does it make sense?
4. Create and plot ROC curves for your classifiers from 3. (2-class problem only). Interpret them.

Extra exercise for the curious:
The target variable ranges from 0 to 4, indicating the number of vessels affected.
Explore ways to model this ordered outcome instead of reducing it to a binary task, without assuming that the target is continuous. Does encoding this assumption improve the results?